

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

```

1 #include <stdio.h>
2
3 void quickSort(int a[], int low, int high) {
4     if (low < high) {
5         int i = low;
6         int j = high;
7         int pivot = a[low];
8
9         while (i < j) {
10             while (a[i] <= pivot && i < high)
11                 i++;
12
13             while (a[j] > pivot)
14                 j--;
15
16             if (i < j) {
17                 int temp = a[i];
18                 a[i] = a[j];
19                 a[j] = temp;
20             }
21         }
22
23         // Place pivot in correct position
24         int temp = a[low];
25         a[low] = a[j];
26         a[j] = temp;
27
28         quickSort(a, low, j - 1);
29         quickSort(a, j + 1, high);
30     }
31 }
32
33 int main() {
34     int n;
35     scanf("%d", &n);
36
37     int a[n];
38
39     for (int i = 0; i < n; i++) {
40         scanf("%d", &a[i]);
41     }
42
43     quickSort(a, 0, n - 1);
44
45     for (int i = 0; i < n; i++) {
46         printf("%d", a[i]);
47         if (i < n - 1)
48             printf(" ");
49     }
50
51     return 0;
52 }

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓